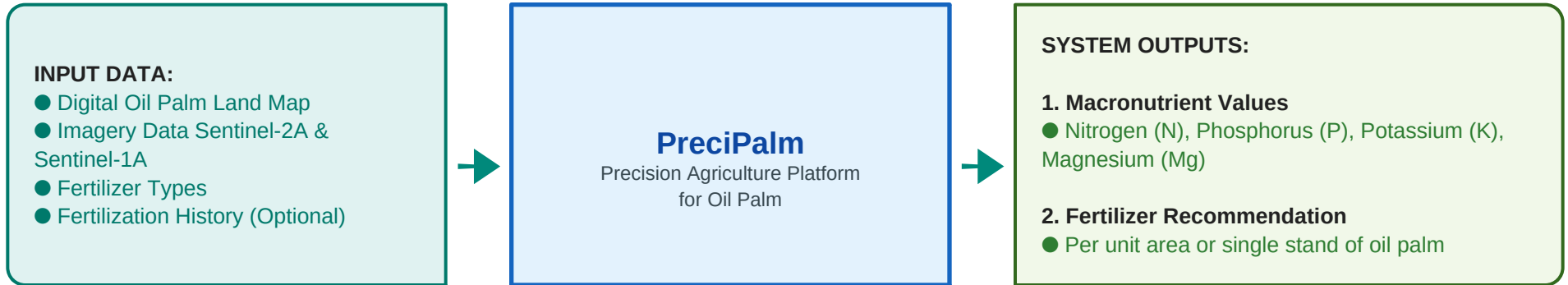


Smart & Precision Agriculture

PreciPalm© — Satellite-Based Nutrient Detection & Fertilizer Recommendation System for Oil Palm

Collaborative Research: IPB University & Pupuk Kalimantan Timur (PKT)



Key Achievements & Validations:

- **Registration:** Received Creation Registration Letter from Ministry of Law and Human Rights of Indonesia
- **Award:** Received Award Certification (115 Karya Inovasi Indonesia Paling Perspektif 2023)
- **Readiness Level:** Technology Readiness Level (TRL) : 9 (The highest Rank)
- **Industry Accuracy:** Applied in real industries — **N (95.7%), P (94.46%), K (86.20%), Mg (83.19%)**

Computational Modeling — Imagery Data Acquisition

Sentinel-1A (SAR Radar)

- 2 Sensors
- 9 extracted main & derivative variables of imagery data ($Z_1, Z_2, \dots, Z_9$)

RFR Algorithm:

$$N \approx F_N(Z_1, Z_2, \dots, Z_9)$$

$$P \approx F_P(Z_1, Z_2, \dots, Z_9)$$

$$K \approx F_K(Z_1, Z_2, \dots, Z_9)$$

Output: Macronutrient values (N, P, K)

Sentinel-2A (Multispectral Optical)

- 12 Spectral Sensors
- 12 extracted main variables of imagery data ($X_1, X_2, \dots, X_{12}$)

RFR Algorithm:

$$N \approx F_N(X_1, X_2, \dots, X_{12})$$

$$P \approx F_P(X_1, X_2, \dots, X_{12})$$

$$K \approx F_K(X_1, X_2, \dots, X_{12})$$

Output: Macronutrient values (N, P, K)

Intelligent and Precision Computing Process in PreciPalm

Cloud Masking & Data Transfer Architecture:

1. **Sentinel-1A Image Data:** Machine Learning model based on Sentinel-1A (Radar penetrating cloud cover).
2. **Sentinel-2A Image Data:** Machine Learning model based on Sentinel-2A (High optical resolution, affected by cloud cover).
3. **Cloud Transfer Process:** Transfer of Sentinel-1A based nutrient status to Sentinel-2A based nutrient status area obscured by clouds.
4. **Result: Cloud-Free Hara Status Map** → Generates accurate **Nutrient Status Map (N, P, K, Mg) & Recommended Dosage**.

High Performance Computing (HPC) Implementation:

- **Parallel Processing (Threading):** Used for intensive cloud masking and feature extraction.
- **Sequential Processing:** Used for final nutrient map rendering and dosage calculation.

PreciPalm End-to-End Machine Learning & Spatial Modeling Flowchart

1. DATA INPUT

- Hasil observasi hara mikro
- Sentinel-2 spektral band
- Koordinat geografis



2. DATA PRE-PROCESSING

- Harmonisasi band (resolusi 10m)
- Inspeksi distribusi nilai data sampel
- Penanganan nilai invalid
- Outlier screening



3. FEATURE ENGINEERING & REDUNDANCY

- Indeks vegetasi & transformasi non-linier
- Analisis multikolinieriti
- Seleksi fitur korelasi tinggi



4. SPATIAL PARTITIONING & K-FOLD CV

- Generasi blok/grid koordinat sampel
- Memberikan ID blok spasial
- Pengelompokan & K-Fold CV tanpa leakage

7. BASE LEARNER TRAINING

Menggunakan:

- XGBoost
- LightGBM
- CatBoost
- Random Forest, dll



8. STACKED ENSEMBLE LEARNING

Menggunakan:

- Meta-learner: BayesianRidge, RidgeCV, atau lainnya



9. SPATIAL RESIDUAL MODELING

Training model residual:

- Gaussian Process Regression / Ordinary Kriging



10. FINAL PREDICTION & DIAGNOSTICS

Prediksi = Prediksi Model Ensemble + Prediksi Residu Spasial

- Uji autokorelasi spasial: Moran's I
- Evaluasi: R², RMSE, RPD, RPIQ
- Serialisasi model akhir